DILPREET KAUR
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SUMMARY

B.Tech IT student building and deploying AI/ML applications, including multi-agent systems, agentic RAG and NLP pipelines, with live demos for each project. Completed an ML internship at NIC working on transformer-based log analysis. Looking for AI, ML and AI engineering roles.

EDUCATION

Guru Tegh Bahadur Institute of Technology, Delhi 2023–2027
B.Tech in Information Technology, CGPA: 8.38

SS Mota Singh School, Delhi 2022–2023
Class X: 86%, Class XII: 80%

EXPERIENCE

Machine Learning Intern | National Informatics Centre (NIC) Jun 2025 – Aug 2025

- Built Gradio and Streamlit interfaces that present model outputs clearly for users without coding expertise.
- Used T5 and BERT to summarize and classify cybersecurity logs, turning dense technical data into short, readable summaries for faster triage.
- Cleaned and organized large log datasets (Pandas, regex) into a labeled structure covering 7+ security event categories.
- Fine-tuned a T5-small model for security-log summarization on 500 labeled logs, reaching 0.91 ROUGE-1/ROUGE-L on validation.
- Evaluated model performance, identified data-quality and generalization issues, and explored improvements.
- Delivered an end-to-end transformer-based log analysis workflow, from raw data preprocessing to model evaluation and a user-facing interface.

PROJECTS

IT Support Copilot | Agentic RAG System *Live Demo*

- Built an agentic RAG copilot that routes questions between direct chat, a private knowledge base and web search, using evidence grading, query rewriting and a retry guard to avoid guessing.
- Added admin-protected document upload (PDF, DOCX, TXT, MD), a decision trace and audit logging.
- **Tools:** Python, LangGraph, FastAPI, Pinecone, Groq, Tavily, SQLite, Streamlit

AI Travel Agent | Multi-Agent Travel Planning System *Live Demo*

- Built and deployed a multi-agent travel planner that coordinates flights, hotels and itineraries into one personalized response for the user.
- Added persistent conversational memory (PostgreSQL) so the planner remembers user requirements across multi-turn chats.
- Integrated LLM and search APIs (LangGraph, Tavily, Groq) behind one Streamlit interface.
- **Tools:** Python, LangGraph, LangChain, Groq, PostgreSQL, Tavily, Streamlit

AI-Powered Deep Packet Inspection & Threat Detection *Live Demo*

- Built a Streamlit dashboard that presents traffic insights and threat scores in a clear, actionable format.
- Trained Random Forest models on 32 engineered features to classify applications and detect threats across 5 traffic classes.
- Developed a multithreaded C++ analyzer for PCAP network traffic, identifying applications via TLS SNI and HTTP Host.
- **Tools:** C++, Python, Scikit-learn, Pandas, Streamlit

SKILLS

- **AI & LLM Engineering:** Agentic RAG, Multi-Agent Systems, LLM Applications, Prompt Engineering, LangGraph, LangChain, Pinecone (Vector DB), Embeddings, Groq API, Tavily API
- **Machine Learning & NLP:** Machine Learning, NLP, Transformers (T5, BERT), Hugging Face, Random Forest, LightGBM, Scikit-learn, Feature Engineering, Model Evaluation (ROUGE)
- **Backend & Deployment:** FastAPI, REST APIs, Streamlit, Gradio, Git, GitHub, MLflow
- **Programming & Data:** Python, C++, SQL, PostgreSQL, SQLite, Pandas, NumPy
- **Core CS:** Data Structures, DBMS, Operating Systems, Computer Networks, OOP

ACHIEVEMENTS & CERTIFICATIONS

- Secured 1st Position at Perfect Pitch 2.0 Ideathon (E-Cell GTBIT x NASSCOM Foundation).
- Certifications: Microsoft Generative AI, IBM Machine Learning with Python.